



Predicting Fall Likelihood in Patients Using Sway Index: A Comprehensive Study on Biometric Factors

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Abstract

Falls are the second leading cause of unintentional injury deaths, after road traffic injuries. The cause is a correlation between health conditions, body structure and function, environmental and personal factors. Examining balance using artificial intelligence algorithms on a large diverse dataset collected in a clinical setting and to provide insight into the underlying mechanisms of falls in older adults is the core objective of this research project. The goal is to shift the focus from reactive treatment to prevention.

Future Goals

- In our most recent Dataset, we acquired a variable that essentially confirms if a patient experienced a fall.
 - This variable could expedite the programming process as we can directly visualize the relationship between various datum and the patient's likelihood of experiencing a fall.
- We plan to use this in addition to other factors like shoes on or shoes off to increase our model's accuracy.
- Identify a relationship between independent and dependent variables
- Test several other models on the new dataset to see if any of them are more accurate.
- Publication of a paper that documents the development process of the model

Methodology

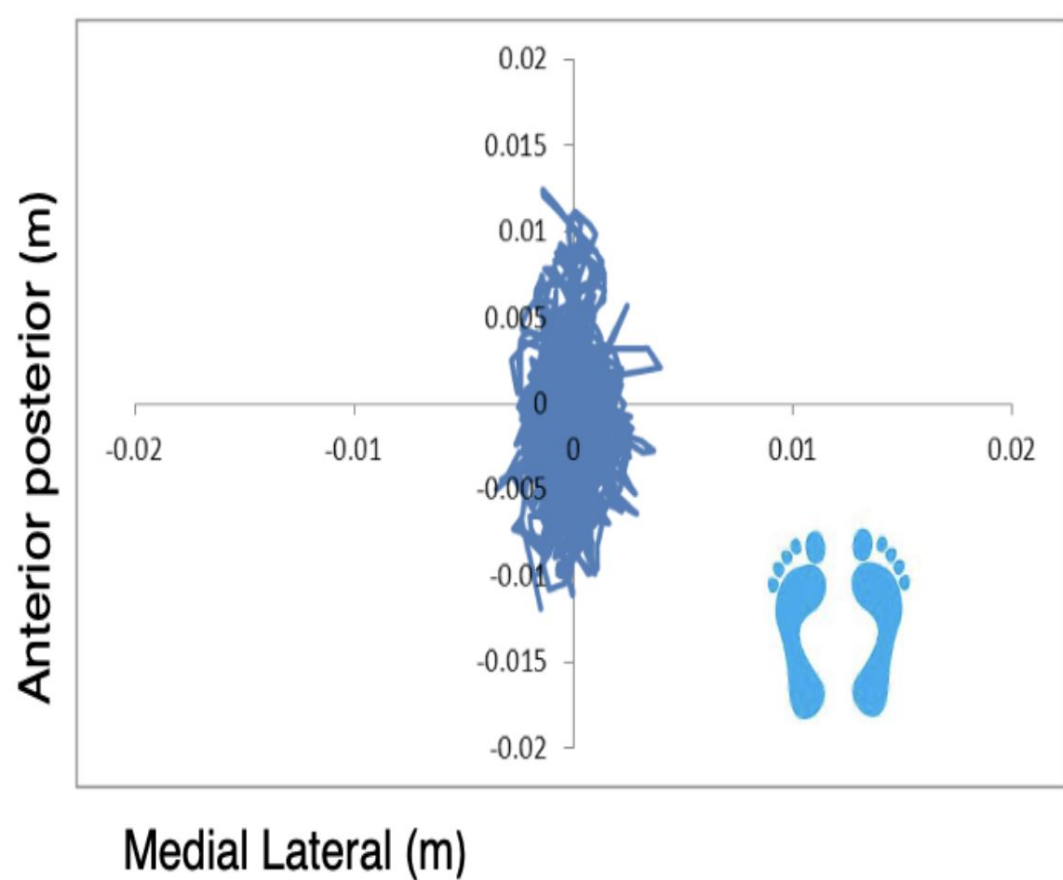
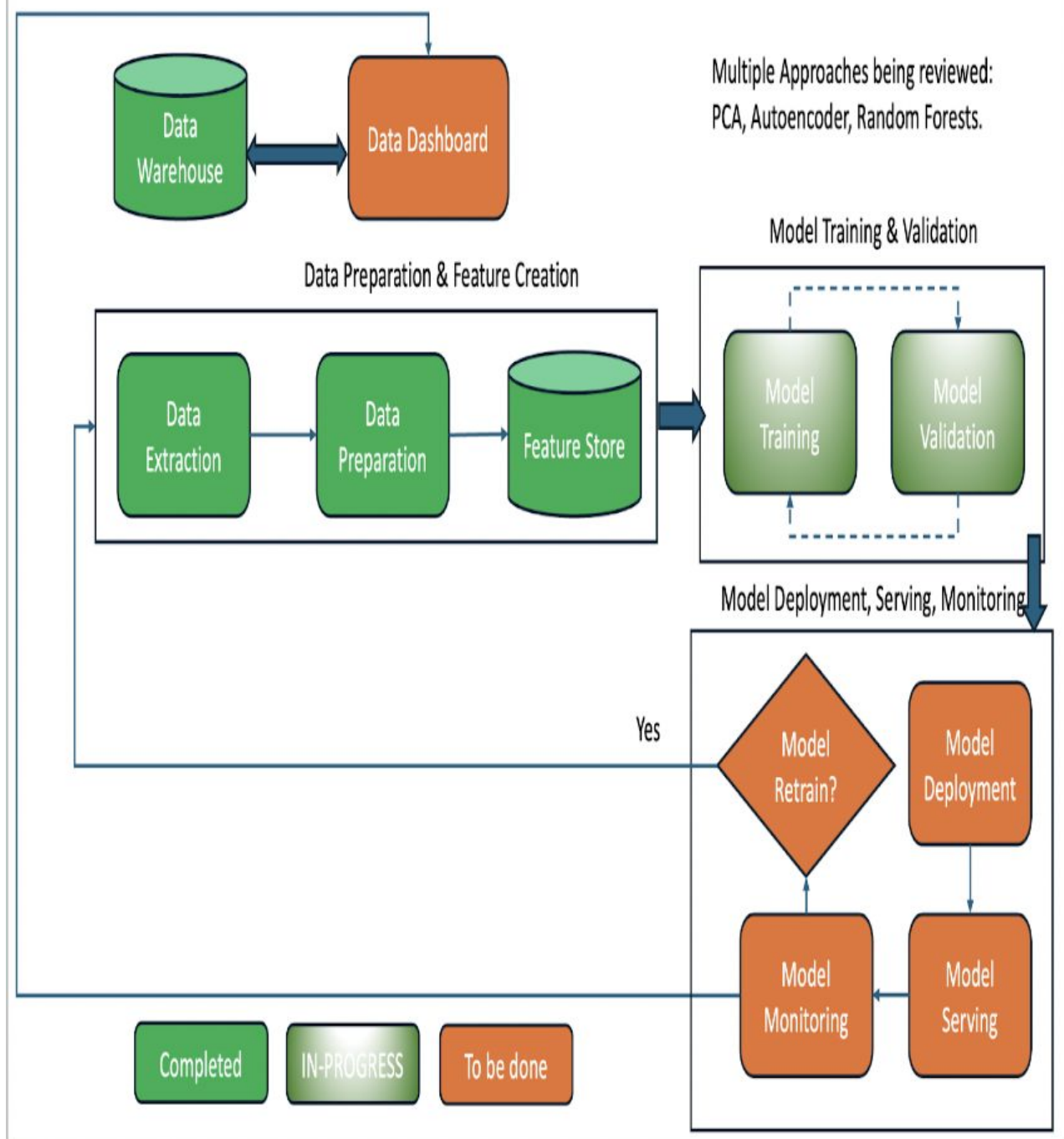


Fig. 1 Balance plot during quiet standing.



- This methodology has been applied to several models including Autoencoders, Random Forest Regressor, and Neural Networks.
- We dropped several values that were insignificant or misleading.
- We will treat four data values as independent and consider the rest to be dependent.

Results for Old Data

- * As of right now, the Neural Network we have programmed can predict Sway Index with an r2 score of 93%.
- * As seen in figures 1 and 2, there is a high correlation between the Sway Index and other clinical data.
- * Due to the uniqueness of sway index in the dataset and the lack of pattern a PCA/Autoencoder approach did not yield an accurate model
- * RNN seems to have the highest accuracy out of all of the models.
- * Utilization of KNN seems to have a slightly lower accuracy.

Motivation

- * Falls are a significant and major health problem that result in serious injury. Each year 684,000 individuals die from falls globally.
- * If not fatal, many falls require medical attention and result in more years lived with disability and a decrease in life expectancy.

Results for New Data

- * We recently acquired the new dataset, so not much progress has been made.
- * We have set up tools like a correlation matrix to identify relationships within the dataset.

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References

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